A Multi-Resolution Quantum Resonance Engine with Fractional Memory Integration and Non-Linear Van der Pol Stability Analysis

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Abstract

This paper formalizes a novel multi-resolution algorithmic framework designed to isolate high-frequency spectral synchronization and predict structural state transitions in highly volatile financial asset feeds. By pairing a restricted GARCH$(1,1)$ volatility filter with a zero-heap multi-scale Rescaled Range (R/S) Hurst calculator, we decouple stationary structural elements from localized fractional memory accumulation. When systemic herding breaches predefined limits, the engine maps the boundary parameters into a non-linear, state-dependent Van der Pol oscillator system. Forward projection via an explicit 4th-Order Runge-Kutta scheme maps the divergence of instantaneous total energy to yield precise, time-bounded collapse predictions before phase space breakdown occurs.

1 Introduction

Modern electronic marketplaces operate as complex adaptive networks driven by multi-scale information feedback loops, structural liquidity patterns, and sudden regime transitions. Under normal operating parameters, price fluctuations exhibit localized diffusive characteristics that align with random walk behavior. However, when distressed conditions or systemic shocks arise, the system undergoes severe structural changes. In these cases, behavioral alignment and recursive feedback loops amplify micro-level correlations into market-wide synchronization.

Standard risk monitoring tools regularly fail to capture these non-linear transformations because they assume stationary statistical traits or treat volatility as a standard mean-reverting process. This paper presents an alternative framework: the *Quant Resonance Engine* (QRE). The QRE views financial returns through the lens of non-linear mechanics and signal analysis. By extracting the spectral fingerprints of localized price feeds, our architecture bridges the gap between historical volatility measures and real-time phase space tracking. This enables practitioners to identify and monitor the early signals of impending systemic breakdowns.

2 Literature Review

The formal study of systemic market instability spans multiple quantitative fields, heavily drawing on financial econometrics, statistical physics, and non-linear dynamics.

2.1 Volatility Modeling and Residual Whitening

The bedrock of conditional heteroskedasticity modeling originates with the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) framework introduced by Bollerslev. While GARCH structures are excellent at capturing short-term volatility clustering and leptokurtic distribution properties, standard residual diagnostic models treat white noise anomalies as independent, identically distributed (*i.i.d.*) components. The framework presented here expands on this by analyzing these filtered residuals as structural indicators that reflect the underlying thermodynamic state of the market.

2.2 Fractional Brownian Motion and Persistent Memory

The use of long-range dependence metrics within financial assets traces back to Mandelbrot's application of Hurst's rescaled range (R/S) mechanics to asset return sequences. A Hurst exponent value $H > 0.50$ implies a persistent, trend-reinforcing memory structure often linked to investor herding and delayed information processing. Capturing this persistence dynamically requires localized window partitioning to prevent structural breaks from muddying the memory calculation.

2.3 Non-Linear Oscillators in Financial Phase Space

The mapping of economic time series to non-linear physical systems is supported by a rich body of complex system

literature. Damped harmonic systems have been used to analyze bubble cycles and market corrections. However, simple linear variants cannot capture the autonomous self-excitation and state-dependent damping common to market liquidity crises.

By upgrading our framework to leverage the Van der Pol oscillator framework, we introduce a non-linear relaxation structure. In this setup, the damping parameter alters its sign based on the state variable's magnitude. This mathematical behavior mirrors real-world market dynamics, where low-amplitude fluctuations are sustained by speculative activity while high-amplitude shocks trigger sharp structural corrections.

3 Volatility Filtering and Residual Standardization

Let $r_t \in \mathbb{R}$ denote the discrete log-return of a financial asset at time index $t \in \mathbb{N}$. The conditional local variance sequence σ_t^2 is dynamically evaluated via a restricted GARCH$(1,1)$ parameterization:

$$\sigma_t^2 = \omega + \alpha r_t^2 + \beta \sigma_{t-1}^2 \tag{1}$$

where the structural parameter constraints are defined as $\omega = 5 \times 10^{-6}$, $\alpha = 0.05$, and $\beta = 0.90$. The standardized residual z_t is computed by scaling the empirical return mapping:

$$z_t = \frac{r_t}{\sigma_t} \tag{2}$$

The sequence z_t is continuously ingested into a static sliding allocation ring buffer $\mathcal{B}_t$ with a maximum capacity $W = 256$:

$$\mathcal{B}_t = \{z_{t-n+1}, z_{t-n+2}, \ldots, z_t\}, \quad n = \min(t, W) \tag{3}$$

4 Multi-Scale Range Split and Fractional Memory Estimation

To track long-range persistence within the whitened residuals without heap allocations, a multi-scale Rescaled Range (R/S) algorithm estimates the local Hurst exponent H_t. For an active buffer length $n \geq 32$, we evaluate the global mean μ and empirical variance σ_z^2:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} \mathcal{B}_t[i], \quad \sigma_z^2 = \frac{1}{n-1}\sum_{i=1}^{n} (\mathcal{B}_t[i] - \mu)^2 \tag{4}$$

The full-scale accumulation deviation provides the global rescaled range $(R/S)_n$:

$$R_n = \max_{1 \leq k \leq n} \sum_{i=1}^{k} (\mathcal{B}_t[i] - \mu) - \min_{1 \leq k \leq n} \sum_{i=1}^{k} (\mathcal{B}_t[i] - \mu) \tag{5}$$

$$\left(\frac{R}{S}\right)_n = \frac{R_n}{\sigma_z} \tag{6}$$

To calculate the scaling slope across a dyadic split, a sub-window of size $n_h = \lfloor n/2 \rfloor$ is evaluated on the right half of the buffer using localized mean μ_h and standard deviation σ_h:

$$\mu_h = \frac{1}{n_h}\sum_{i=n_h+1}^{n} \mathcal{B}_t[i], \quad \sigma_h = \sqrt{\frac{1}{n_h - 1}\sum_{i=n_h+1}^{n} (\mathcal{B}_t[i] - \mu_h)^2} \tag{7}$$

$$R_{n_h} = \max_{1 \leq k \leq n_h} \sum_{i=n_h+1}^{n_h+k} (\mathcal{B}_t[i] - \mu_h) - \min_{1 \leq k \leq n_h} \sum_{i=n_h+1}^{n_h+k} (\mathcal{B}_t[i] - \mu_h) \tag{8}$$

The fractional dimensional scaling slope determines the bounded Hurst exponent H:

$$H = \max\left(0.01, \min\left(0.99, \frac{\ln(R_n/\sigma_z) - \ln(R_{n_h}/\sigma_h)}{\ln(2)}\right)\right) \tag{9}$$

5 Spectral Harmonic Resonance Mapping

The engine maps high-frequency spectral density by passing $\mathcal{B}_t$ into a Discrete Fourier Transform (DFT). The discrete power spectrum $P(k)$ is surveyed over the sub-Nyquist search space $k \in [1, \lfloor n/2 \rfloor)$:

$$P(k) = \left| \sum_{j=0}^{n-1} \mathcal{B}_t[j+1] \cdot \exp\left(-i\frac{2\pi k j}{n}\right) \right|^2 \tag{10}$$

The peak energy bin $k_{\text{peak}} = \arg\max_k P(k)$ establishes the localized structural resonance frequency ω_0 given sampling frequency f_s:

$$f_{\text{hz}} = \frac{k_{\text{peak}} \cdot f_s}{n}, \quad \omega_0 = 2\pi f_{\text{hz}} \tag{11}$$

We compute the system Quality Factor (Q) and corresponding non-linear base damping coefficient μ_{vdp}:

$$Q = \begin{cases} \frac{\omega_0}{\sigma_z} & \text{if } \sigma_z^2 > 10^{-8} \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

$$\mu_{\text{vdp}} = \begin{cases} \frac{\omega_0}{2Q} & \text{if } Q > 0 \\ \omega_0 & \text{otherwise} \end{cases} \tag{13}$$

6 Van der Pol Non-Linear Energy Projection and Collapse Forecasting

A projection sequence triggers if and only if the structural quality factor and long-range memory simultaneously breach critical limits:

$$\text{Trigger} \iff \Big(Q \geq Q_{\text{crit}}\Big) \wedge \Big(H \geq H_{\text{limit}}\Big) \wedge \Big(\omega_0 > 0\Big) \tag{14}$$

where $Q_{\text{crit}} = 2.5$ and $H_{\text{limit}} = 0.55$. Upon activation, the state vector is mapped into a non-linear, self-excited Van der Pol differential equation:

$$\frac{d^2x}{dt^2} - \mu_{\text{vdp}}(1 - x^2)\frac{dx}{dt} + \omega_0^2 x = 0 \tag{15}$$

Let $\mathbf{u} = [x, v]^T$, where the state conditions match $x(0) = \mathcal{B}_t[n]$ and velocity $v(0) = (\mathcal{B}_t[n] - \mathcal{B}_t[n-1]) \cdot f_s$. The system is integrated forward for steps $m \leq m_{\max} = 120$ via an explicit Runge-Kutta 4th-Order scheme with step-size $\Delta t = 1/f_s$:

$$\mathbf{u}_{m+1} = \mathbf{u}_m + \frac{\Delta t}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right) \tag{16}$$

where the vector field $\mathbf{f}(x, v)$ adapts to the Van der Pol model:

$$\mathbf{f}(x, v) = \begin{bmatrix} v \\ \mu_{\text{vdp}}(1 - x^2)v - \omega_0^2 x \end{bmatrix} \tag{17}$$

The standard derivative coefficients are iteratively evaluated:

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{u}_m) \tag{18}$$

$$\mathbf{k}_2 = \mathbf{f}\left(\mathbf{u}_m + \frac{\Delta t}{2}\mathbf{k}_1\right) \tag{19}$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{u}_m + \frac{\Delta t}{2}\mathbf{k}_2\right) \tag{20}$$

$$\mathbf{k}_4 = \mathbf{f}\left(\mathbf{u}_m + \Delta t\mathbf{k}_3\right) \tag{21}$$

The instantaneous system energy is checked against geometric boundary layers:

$$E_m = \frac{1}{2}v_m^2 + \frac{1}{2}\omega_0^2 x_m^2 \tag{22}$$

The predicted time-to-collapse T_c registers at the first step m satisfying:

$$T_c = m \cdot \Delta t \tag{23}$$

where

$$m = \min\left\{m \mid E_m > 3.5 \cdot E_0 \ \vee \ |x_m| > 4.0\right\} \tag{24}$$

If the threshold conditions are not reached within the forecast horizon $m_{\max}$, the current state is classified as **Stable**.

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